

Transition-State Damping as a Mechanistic Descriptor of Rate Turnover in Chemical Reactivity

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Abstract

Chemical rate theory has a well-developed thermodynamic axis but lacks a unified framework for the dynamical axis: the extent to which a reactive coordinate commits productively rather than re-crossing. This research shows that dynamical commitment near a transition state is governed by a dimensionless damping ratio, $\chi = \Gamma_{\text{eff}}/(2\Omega_{\text{TS}})$, which separates oscillatory recrossing from overdamped sluggishness and is maximized at $\chi \approx 1$. Established theories of barrier crossing in electron transfer, proton-coupled electron transfer, and heterogeneous catalysis reduce locally to this same dynamical structure despite arising from distinct microscopic mechanisms. Three calibrated estimator routes connect χ to spectroscopic and computational observables with defined validity limits. Worked examples illustrate solvent-dependent rate turnover in TEMPO radical electron transfer and K-promoter effects in Fe(111)-catalyzed N_2 activation. Eight falsification tests are specified; the framework is refuted if the rate turnover predicted at $\chi \approx 1$ is absent when thermodynamic parameters are held fixed. This dynamical descriptor introduces an axis orthogonal to thermodynamic rate theory, enabling quantitative interpretation of recrossing, catalytic outliers, and mechanism transitions from a single dimensionless parameter.

Reader’s guide. Sections 1 and 2 establish the theoretical framework and estimator routes. Readers primarily interested in applications may proceed directly to Section 4 and consult the tables in earlier sections as needed. The validity boundaries in Section 5 should be reviewed before applying the framework to a new system.

Introduction

Rate theory has a well-understood thermodynamic axis: transition-state theory connects barrier height and driving force to rate constants, and Brønsted–Evans–Polanyi (BEP) scaling relations organize these across reaction families. What thermodynamic theory does not address is whether the reactive coordinate, having arrived at the transition state, actually completes the transformation or recrosses back. This gap is experimentally real. Identical donor–acceptor pairs with the same ΔG° and $\Delta G^\ddagger$ produce dramatically different charge-separated-state populations across solvents. PCET systems shift between stepwise and concerted mechanisms under changes in environmental control parameters such as solvent, composition, or pressure, even when the thermodynamic landscape remains broadly similar. Fe(111) outperforms Fe(110) in N_2 activation despite comparable binding energies. In each case, the missing variable is dynamical commitment.

The ratio $\chi = \Gamma_{\text{eff}}/(2\Omega_{\text{TS}})$, defined as the total projected dissipation rate divided by twice the barrier-top curvature frequency, was established in prior work²⁹ as the parameter governing solution character near any transition state: below $\chi = 1$ solutions are oscillatory and recrossing occurs; above it they decay monotonically with increasing sluggishness; at $\chi = 1$ the rate of productive commitment is fastest. The present paper operationalizes this ratio through three calibrated estimator routes connecting χ to standard laboratory observables, with interpretation protocols, worked examples, and explicit validity boundaries,

making χ a routinely computable and falsifiable descriptor alongside thermodynamic rate theory. This ratio is termed here the Critical Chemical Equivalence (CCE) parameter.

Section 1 establishes the generator and domain mapping. Section 2 derives the three estimator routes. Section 3 provides the quantitative interpretation protocol. Section 4 develops the design levers with domain-specific worked examples and experimental predictions. Section 5 defines the validity boundaries. Derivations, bias analysis, and the complete falsification protocol are in the Supplementary Information.

1 The Dynamical Ratio: Generator, Boundary, and Domain Mapping

Near any transition state, projection of the full potential energy surface onto the committor-dominant coordinate reduces locally to a second-order damped oscillator:

$$\ddot{q} + \Gamma_{\text{eff}} \dot{q} + \Omega_{\text{TS}}^2 q = \xi(t) \quad (1)$$

where Ω_{TS} is the magnitude of the curvature frequency at the barrier top and Γ_{eff} is the total projected dissipation from all channels: solvent friction, vibrational coupling, or electronic hybridization with a metal surface, depending on the domain. The reactive coordinate at a saddle point has inverted curvature; Eq. (1) describes the projected dynamics in the Kramers/PGH formulation where Ω_{TS} enters through the reactive-mode frequency magnitude. The qualitative character of the solution is set by the discriminant $\Delta = \Gamma_{\text{eff}}^2 - 4\Omega_{\text{TS}}^2$, which changes sign at exactly one point:

$$\chi_{\text{TS}} \equiv \frac{\Gamma_{\text{eff}}}{2\Omega_{\text{TS}}} = 1 \quad (2)$$

Below this boundary ($\chi < 1$) solutions are oscillatory and recrossing occurs. Above it ($\chi > 1$) solutions decay monotonically but with increasing sluggishness. At $\chi = 1$, monotonic commitment is fastest; the generator reaches a second-order exceptional point¹ where eigenvalues coalesce and the Green’s function becomes polynomial-exponential (Supplementary §1.1). This is a mathematical property of the generator Eq. (1), not a chemical model.

Three independently derived theoretical frameworks (Kramers (1940)² (reviewed comprehensively in Hänggi et al. 1990³), Zusman (1980)⁴, and Newns–Anderson (1969)⁵) reduce locally to this same generator through different microscopic mechanisms, as summarized in Table 1. χ is the single ratio that organizes all three.¹

¹The physical origin of Γ_{eff} differs across domains: nuclear Langevin friction in Kramers, dielectric solvent relaxation in Zusman, electronic hybridization width in Newns–Anderson. The mathematical structure of the generator is identical. The claim is structural reduction, not ontological identity.

Domain	Γ_{eff} is	Ω_{TS} is	χ governs
ET / Kramers	Solvent friction ζ or $1/\tau_L$	Barrier curvature ω_b or reorganization frequency	Friction-controlled recrossing vs. commitment; interacts with electronic adiabaticity (Zusman <i>g</i>)
PCET	Coupled solvent and vibronic damping Γ_{eff}	Frequency of the combined PT-ET coordinate	Stepwise vs. concerted mechanism selection
Catalysis (Newns-Anderson)	Electronic friction from d-band hybridization γ_{NN}	Adsorbate stretch frequency, softened by π^* -backdonation	Bond activation efficiency; BEP outliers

Table 1: Domain instantiations of the generator Eq. (1). The physical origin of Γ_{eff} differs across domains; the mathematical structure is identical.

The reduction is structural: each framework’s linearized committor-projected dynamics is governed by the same second-order operator regardless of microscopic origin. No assumptions of shared ontology are made.

The three dynamical regimes and their chemical consequences are given in Table 2. Two points about these regimes deserve emphasis. First, the dynamical optimum is a window $\chi \in [0.8, 1.2]$, not the point $\chi = 1.000$ exactly. The Pollak–Grabert–Hänggi (PGH) combined rate ⁶ stays within 5% of its peak throughout this window for thermally activated systems with $V^\ddagger/k_B T \geq 8$ (Supplementary §1.2). Second, all three estimator routes (Section 2) yield χ_{GS} , a lower bound on χ_{TS} : bond softening and increasing π^* -backdonation on approach to the transition-state geometry decrease Ω_{TS} while leaving Γ_{eff} unchanged or larger, raising χ along the reaction path. A ground-state diagnosis of $\chi_{\text{GS}} \approx 0.6$ is therefore consistent with $\chi_{\text{TS}} \approx 1$ if the reactive bond softens significantly toward the saddle point. Ground-state estimates should never be treated as upper bounds on χ_{TS} without independent transition-state geometry data or reaction-path normal-mode analysis.

χ	Regime	Coordinate behavior	Chemical consequence
< 0.8	Underdamped	Oscillatory; recrossing probable	Back-transfer; poor yield; weak bond activation
$0.8\text{--}1.2$	Near-critical	Fastest monotonic commitment	Maximum ET yield; concerted PCET; peak catalytic TOF
> 1.2	Overdamped	Monotonic but sluggish	Solvent-controlled ET; inhibited PCET; poisoned catalyst

Table 2: The three dynamical regimes. The window $[0.8, 1.2]$ captures $\geq 95\%$ of the available dynamical efficiency under physically relevant barrier heights (Supplementary §1.2).

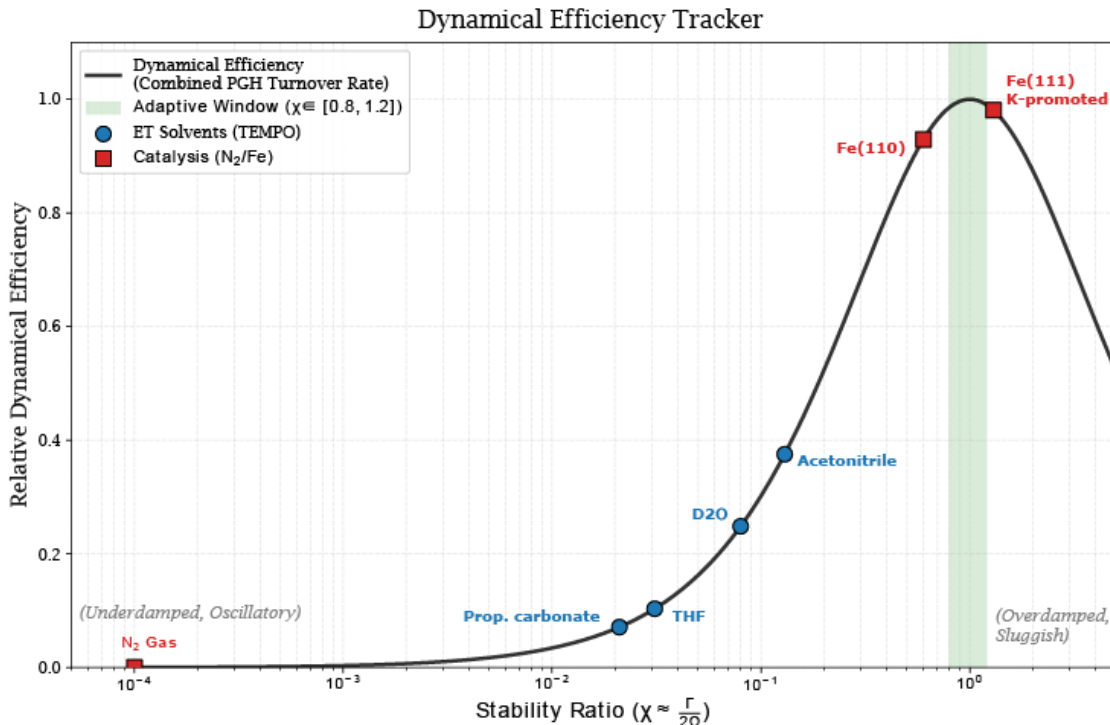


Figure 1: Dynamical efficiency across domains. Normalized PGH combined rate $\eta(\chi) = k(\chi)/k_{\max}$ as a function of the damping ratio $\chi = \Gamma_{\text{eff}}/(2\Omega_{\text{TS}})$, plotted on a log scale. Shaded band: near-critical window $\chi \in [0.8, 1.2]$, where $\eta \geq 0.95$. Blue circles: TEMPO radical across solvents of increasing τ_L (Route B, relative χ ; m_{eff} cancels within the series). Red squares: N_2/Fe surfaces (Route C; all values are lower bounds on χ_{TS}). N_2 gas phase sits at $\chi \sim 10^{-5}$, off-axis to the left.

The framework applies only when the transition-state region is locally harmonic in the reactive coordinate. Strongly anharmonic transition states invalidate the generator Eq. (1); the committor-projected potential should be verified to be parabolic before applying χ . When multiple vibrational modes couple comparably to the reactive coordinate, the scalar χ must be replaced by the mode-projected effective quantity χ_{eff} (Supplementary §6.1); a single χ is adequate only when one mode carries more than 70% of the reactive weight.

2 Estimator Routes: Three Paths from Observables to χ

Because χ_{TS} is defined at the transition state, it cannot be measured directly. Three estimator routes connect it to laboratory observables, each with a defined domain of validity, a defined bias, and explicit conditions for when it is inapplicable. Routes should not be mixed within a single comparison.

Route	Inputs	Formula	Applies when	Bias and limit
A: Spec-troscopic	$\Delta\tilde{\nu}_L, \tilde{\nu}$ (cm^{-1})	$\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$	Operando IR or Raman; surface catalysis	Valid only when Lorentzian fraction $> 50\%$; linewidth must be coverage-independent; cross-check with T_1 if available
B: Solvent relax-ation	τ_L (ps), λ (eV), ΔQ ($\text{\AA}$), m_{eff} (amu)	$\Omega = \sqrt{2\lambda/(m_{\text{eff}}\Delta Q^2)}$; $\chi_{\text{ET}} = 1/(2\tau_L\Omega)$	Electron transfer; PCET; solvent-controlled kinetics	m_{eff} is uncertain by factor 3–10; relative χ eliminates this uncertainty and is the recommended quantity
C: DFT friction	γ_{NN} (eV), $\tilde{\nu}$ (cm^{-1})	$\chi_{\text{DFT}} = (\Gamma_{\text{int}} + \gamma_{NN})/(2hc\tilde{\nu})$; $hc = 1.240 \times 10^{-4} \text{ eV}\cdot\text{cm}$	Surface catalysis with DFT electronic friction available	Mean-field DFT must be valid; 3d ferromagnets may require DFT+ U correction

Table 3: Three estimator routes for χ_{TS} . All three are lower bounds on χ_{TS} in systems where the bond softens toward the transition state.

Route A: Spectroscopic estimator

Route A uses vibrational linewidth as a proxy for the dissipation rate. The true dissipation rate is $\Gamma = 1/T_1$ (vibrational population lifetime); the measured IR linewidth reflects T_2 coherence decay via $1/T_2 = 1/(2T_1) + 1/T_2^*$, where T_2^* encodes pure dephasing and inhomogeneous broadening. Route A gives $\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$, which equals the true χ_{TS} only when $T_2 \approx 2T_1$, i.e., when homogeneous lifetime broadening dominates.

The raw FWHM from spectrometer software must not be used. Every peak must be fit to a Voigt profile, and only the Lorentzian component $\Delta\tilde{\nu}_L$ extracted. When the Gaussian component exceeds the Lorentzian component, the sample is either too warm or too inhomogeneous, and the linewidth no longer reflects the dissipation channel; the estimate will be spuriously high, and the measurement should be repeated at lower temperature or lower adsorbate coverage. The measurement procedure is: (1) acquire operando IR under steady-state turnover conditions; (2) fit a Voigt profile and extract $\Delta\tilde{\nu}_L$ separately from $\Delta\tilde{\nu}_G$; (3) confirm Lorentzian fraction $> 50\%$ and coverage-independence of the linewidth; (4) compute $\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu}_{\text{peak}})$; (5) if T_1 data are available from pump–probe vibrational spectroscopy, verify that $\Gamma_{\text{true}} = 1/T_1$ gives consistent χ within a factor of two. Lorentzian lineshape is necessary but not sufficient proof of homogeneous broadening; coverage independence and T_1 cross-checks are essential.

Route B: Solvent-relaxation estimator

Route B connects χ to the longitudinal dielectric relaxation time τ_L , which sets the timescale of solvent friction in Kramers and Zusman theories. The formula requires m_{eff} , the effective mass of the collective solvation mode, which is not obtainable from first principles. Literature estimates for m_{eff} span 50–500 amu, and because $\chi \propto m_{\text{eff}}^{+1/2}$, a decade of uncertainty in m_{eff} propagates to a factor of $\sqrt{10} \approx 3.2$ in χ . An absolute Route B number reported to two significant figures is not physically meaningful.

The standard Route B quantity is therefore relative χ . When m_{eff} and ΔQ are constant across a solvent series (same donor-acceptor pair, varying solvent), m_{eff} cancels exactly:

$$\frac{\chi_{\text{ET}}^{(A)}}{\chi_{\text{ET}}^{(B)}} = \frac{\tau_L^{(A)}}{\tau_L^{(B)}} \quad (3)$$

This is the operationally robust quantity. To locate the dynamical optimum, compile k_{et} and τ_L across at least five solvents and plot k_{et} against τ_L . The rate maximum identifies the solvent where $\chi_{\text{ET}} \approx 1$; solvents with shorter τ_L are underdamped, solvents with longer τ_L are overdamped. Eq. (3) maps the entire series relative to that optimum without requiring m_{eff} at any step. To confirm solvent-dynamical control before interpreting the turnover, verify that $k_{\text{et}} \times \tau_L$ is approximately constant across the series. If a bracketed absolute estimate is required for screening, compute χ_{ET} for $m_{\text{eff}} \in \{50, 100, 200, 500\}$ amu and report the full range; a regime assignment is reliable only if the entire bracket falls within a single regime.

Route C: DFT electronic friction estimator

Route C is the highest-fidelity estimator when electronic friction dominates, as in chemisorption on metal surfaces. The mode-projected electronic friction tensor γ_{NN} is obtained from density-functional perturbation theory¹³. Then:

$$\chi_{\text{DFT}} = \frac{\Gamma_{\text{int}} + \Gamma_{\text{cat}}}{2\Omega_{\text{TS}}}, \quad \Gamma_{\text{cat}} = \gamma_{NN}, \quad \Omega_{\text{TS}} = hc\tilde{\nu}_{\text{bond}}$$

As a consistency cross-check, the predicted linewidth $\Delta\tilde{\nu} = \hbar\Gamma_{\text{cat}}/hc$ (in cm^{-1}) should match the Lorentzian component $\Delta\tilde{\nu}_L$ from Route A within a factor of two. Route C computes χ at the ground-state adsorbate geometry and is therefore a lower bound on χ_{TS} : as the reactive bond stretches toward the transition state, π^* -backdonation increases (raising Γ_{cat}) while Ω_{bond} decreases, both effects raising χ above the ground-state value. For quantitative transition-state predictions, γ_{NN} and the harmonic frequency should be computed at the saddle-point geometry along the reaction path. For screening purposes, χ_{DFT} from the ground-state geometry is adequate provided it is interpreted as a lower bound.

3 Quantitative Interpretation and the Regime Decision Protocol

Once a χ estimate is in hand, interpretation requires first verifying that thermodynamic parameters (ΔG° , λ , $\Delta G^\ddagger$) are independently acceptable. χ addresses only the dynamical axis and cannot compensate for an unfavorable barrier or driving force. With thermodynamics confirmed, the regime assignment in Table 4 guides the appropriate response.

χ	Diagnosis	Thermodynamics check	Recommended response
< 0.5	Strongly oscillatory; recrossing dominates	Thermodynamic parameters may be fine; bottleneck is dynamical	Increase Γ_{eff} substantially; confirm system is in scope
$0.5\text{--}0.8$	Oscillatory; commitment partial; back-transfer significant	Verify ΔG° ; if favorable, dynamical adjustment is worthwhile	Moderate increase of Γ_{eff} or decrease of Ω_{TS} ; targeted solvent or promoter change
$0.8\text{--}1.2$	Dynamically optimized	Thermodynamics is now the primary constraint	Do not adjust dynamics; optimize barrier and driving force instead
$1.2\text{--}2.0$	Monotonic but slowing; solvent-controlled or promoter-poisoned	Thermodynamic driving force may be partially consumed by excess friction	Reduce Γ_{eff} or increase Ω_{TS} ; faster solvent; reduce promoter loading
> 2.0	Dynamics severely overdamped; coordinate sluggish	Check for thermodynamic over-binding; may require new scaffold	Major redesign: decouple bath modes, change metal or facet

Table 4: Regime interpretation protocol. The diagnosis assumes thermodynamic parameters have been independently verified.

An underdamped system is not always performing sub-optimally. If the electron-transfer rate is already faster than the downstream process, adjusting χ adds no value. The regime assignment identifies the source of limitation, not a universal performance ranking. The appropriate action in each case depends on whether the dynamical axis is actually the binding constraint.

4 Design Levers, Worked Examples, and Experimental Predictions

The optimization objective is to tune Γ_{eff} and Ω_{TS} such that $\chi_{\text{TS}} \approx 1$ while holding thermodynamic parameters fixed. Changing ΔG° , λ , or $\Delta G^\ddagger$ simultaneously with dynamical variables invalidates a χ -based diagnosis and obscures which axis was responsible for any observed change. Two independent handles exist in every domain: Γ_{eff} can be raised or lowered by modifying the coupling between the reactive coordinate and its bath, and Ω_{TS} can be raised or lowered by modifying the curvature of the potential at the barrier top. In heterogeneous catalysis, a third handle modulates how the reactive coordinate projects onto the friction tensor through mode geometry and surface coordination.

Table 5 summarizes the domain-specific lever assignments. Worked examples follow for each domain.

Domain	Raise χ	Lower χ	Key observable
Electron transfer	More viscous or slower solvent ($\uparrow \tau_L$); larger λ or ΔQ ; heavier donor-acceptor pair	Faster-relaxing solvent ($\downarrow \tau_L$); small- λ structural redesign; less polar medium	k_{et} vs. $1/\tau_L$; slope unity indicates dynamical control
PCET	Increase solvent viscosity; hydrostatic pressure; strengthen H-bond network; H $\rightarrow$ D substitution shifts χ by $\times\sqrt{2}$	Decrease viscosity; reduce pressure; disrupt H-bond donors	KIE and pH-independence as a function of viscosity or pressure
Heterogeneous catalysis	Promoter loading (K, Cs); alloy with more oxophilic metal to increase d-band coupling and electronic friction; π^* -backdonation lowers $\tilde{\nu}$	Reduce promoter; alloy with noble metal to weaken d-band hybridization; increase bond stiffness or partially oxidize surface	Operando $\Delta\tilde{\nu}_L/\tilde{\nu}$ vs. TOF; maximum at $\chi \approx 1$

Table 5: Domain-specific tuning levers. All adjustments should hold ΔG° , λ , and $\Delta G^\ddagger$ as fixed as possible.

Electron transfer: optimizing solvent for maximal yield via relative χ

A donor-acceptor pair with $\lambda = 0.7$ eV and $\Delta G^\circ = -0.3$ eV shows incomplete charge separation across five solvents despite favorable thermodynamics (illustrative of the phenomenology described in Barbara et al. ⁸, Marcus ²⁴, Grampp and Rasmussen ²⁵, and Kumpulainen et al. ⁹). The measured rate constants and longitudinal relaxation times are given in Table 6.

Solvent	τ_L (ps)	k_{et} ($10^8 \text{ M}^{-1}\text{s}^{-1}$)
Acetonitrile	0.20	5.0
Water	0.20	3.5
D ₂ O	0.30	2.8
THF	0.80	1.8
Propylene carbonate	1.20	0.88

Table 6: Electron-transfer rate constants and longitudinal relaxation times across five solvents for a fixed donor-acceptor pair.

The product $k_{\text{et}} \times \tau_L$ takes values 0.10, 0.07, 0.08, 0.14, and 0.11 across the series. These values vary by roughly a factor of two, indicating that while solvent-dynamical control is the dominant contribution, some variation in thermodynamic parameters (λ , ΔG°) across solvents is likely present. The qualitative Route B analysis remains applicable because the τ_L dependence accounts for most of the rate variation; any corrections from solvent-dependent λ shifts would need to be estimated from independent measurements of reorganization energy to sharpen the analysis. A plot of k_{et} against $1/\tau_L$ shows monotonically rising rate with decreasing τ_L , placing all five solvents on the underdamped flank of the Kramers turnover. None has reached the near-critical window. Using Eq. (3), the relative damping spans a factor of $1.20/0.20 = 6\times$

across the series, with the optimum lying at a shorter τ_L than any solvent measured. The framework predicts that solvents or solvent mixtures with $\tau_L < 0.1$ ps (e.g., formamide–water blends or ionic liquids with fast librational modes) will show continuing rate increase followed by a turnover as τ_L falls past the $\chi_{\text{ET}} = 1$ point. The rate decline after the maximum is solvent-controlled ($k \propto 1/\tau_L$) and is mechanistically distinct from simple viscosity effects on diffusion, constituting a falsifiable prediction of the framework.

Proton-coupled electron transfer: mechanism switching via KIE and relative χ

A ruthenium–tyrosine PCET complex^{10,11,12} shows pH-independent rate in water but pH-dependent rate in THF. The question is whether the mechanism shift is dynamical or thermodynamic. Measuring KIE (k_H/k_D) and k_{obs} across four solvents of increasing viscosity at fixed T , λ , and ΔG° allows a Route B diagnosis without requiring knowledge of m_{eff} .

For the proton coordinate, H $\rightarrow$ D substitution increases m_{eff} by a factor of two and therefore raises Ω_{PT} by $\sqrt{2}$, shifting:

$$\chi_{\text{eff}}^D = \frac{\chi_{\text{eff}}^H}{\sqrt{2}} \approx 0.71 \chi_{\text{eff}}^H$$

If $\chi_{\text{eff}}^H \approx 1$ in a given solvent, the deuterated system sits at $\chi_{\text{eff}}^D \approx 0.71$, which is underdamped and stepwise. The KIE trend across the viscosity series encodes the regime traversal: KIE rises as viscosity increases because the system approaches $\chi_{\text{eff}}^H = 1$ from below and concerted PCET emerges; KIE reaches a maximum when $\chi_{\text{eff}}^H \approx 1$; KIE falls at still higher viscosity because both H and D coordinates are now overdamped and the mechanism returns to stepwise, not through tunneling but through sluggishness. pH-independence and absence of detectable intermediates at the KIE maximum confirm the concerted assignment. The KIE maximum is a dynamical signature of crossing the $\chi = 1$ separatrix, not a tunneling signature, and appears in a fully classical framework.

Experimental support for friction-driven mechanism switching comes from Langford and coworkers²⁸, who studied an excited-state Ru-complex PCET system. Under standard concentrations the reaction proceeds stepwise; at high acceptor and buffer concentrations (100 mM MQ⁺, 50 mM phosphate) the mechanism switches to concerted PCET. Pressure measurements serve as the diagnostic, not the forcing lever: the stepwise ET pathway exhibits $\Delta V^\ddagger = -14.5 \pm 0.5 \text{ cm}^3 \text{ mol}^{-1}$, consistent with electrostriction from charge separation, while the CPET pathway gives $\Delta V^\ddagger = -2.9 \pm 3.4 \text{ cm}^3 \text{ mol}^{-1}$, statistically indistinguishable from zero. The near-zero activation volume confirms that simultaneous electron-proton transfer bypasses solvent electrostriction entirely, removing the dominant environmental friction channel. This friction-independence is consistent with the boundary condition $\chi_{\text{eff}} \approx 1$ predicted by the framework, providing an experimental diagnostic compatible with the dynamical interpretation above.

A falsifiable prediction of the framework is that any control parameter that systematically shifts the effective environmental friction, including solvent, composition, or pressure where applicable, should produce a KIE maximum as the system approaches the near-critical window. Failure of the KIE maximum to track the independently identified dynamical crossover at fixed ΔG° and λ would refute the friction-based mechanism assignment (Test 3, Supplementary §4). Diagnostic signatures distinguishing stepwise from concerted PCET are summarized in Table 7.

Observable	Stepwise	Concerted
KIE (k_H/k_D)	2–7	Variable; interpret alongside pH-independence and absence of intermediates
pH dependence	Rate depends on pH	Rate pH-independent
Intermediates	Detectable	None
Pressure response	Large negative $\Delta V^\ddagger$	Near-zero $\Delta V^\ddagger$

Table 7: Diagnostic signatures for PCET mechanism assignment.

Heterogeneous catalysis: operando χ_{spec} as a screening metric for N_2 activation

Operando DRIFTS is available on a series of Fe-based catalysts with varying K-promoter loading. N_2 dissociation is the established rate-determining step for ammonia synthesis on Fe under both industrial and model surface-science conditions^{17,18}, so the dynamical commitment of the N–N bond coordinate is directly kinetically relevant. The objective is to rationalize the TOF-versus-loading curve and predict the over-promotion threshold.

For systems where DFT electronic friction is available^{13,14}, Route C gives direct estimates. Fitting operando IR to a Voigt profile and extracting $\Delta\tilde{\nu}_L$ (the Lorentzian component only) gives Route A estimates. The two should not be applied to the same linewidth: Route A applies to IR data, Route C applies to DFT γ_{NN} values; applying Route A mathematics to DFT linewidths produces incorrect results. Canonical values across the Fe surface series, all via Route C from DFT friction^{15,16}, are given in Table 8.

System	$\tilde{\nu}$ (cm^{-1})	γ_{NN} (eV)	χ_{DFT}
N_2 gas phase	2359	$\sim 10^{-5}$	$\sim 10^{-5}$
$\text{N}_2/\text{Fe}(111)$ unpromoted	~ 2100	~ 0.05	~ 0.10
$\text{N}_2/\text{Fe}(110)$	~ 2000	~ 0.30	~ 0.61
$\text{N}_2/\text{Fe}(111)$ K-promoted	~ 1415	~ 0.45	~ 1.28

All χ_{DFT} are lower bounds on χ_{TS} (ground-state geometry; see Section 2).

Table 8: CCE descriptor across the Fe surface series via Route C.

This framework predicts that TOF maximizes at $\chi_{\text{DFT}} \approx 1$ on a log-linear plot of TOF versus χ ^{19,20,21}. Systems below this value benefit from additional promoter loading; systems above it are over-promoted and lose activity with further loading. Increasing K from zero to the optimal loading simultaneously decreases $\tilde{\nu}$ (via backdonation) and increases $\Delta\tilde{\nu}_L$ (via d-band coupling), driving χ_{spec} toward unity and raising TOF. Exceeding the optimal loading pushes $\chi_{\text{spec}} > 1.2$, and TOF decreases despite continued promoter addition. This prediction is directly falsifiable as Test 7 of the falsification protocol (Supplementary §4): TOF must maximize at finite K loading and decline thereafter.

The Haber–Bosch process conditions (Fe(111), K promoter, 400°C, 200 atm) were arrived at empirically over decades. Retrospectively, the combined effect of promoter loading, elevated pressure, and temperature

correlates with χ_{spec} increasing toward $\mathcal{O}(1)$, simultaneously raising Γ_{cat} via d-band coupling and lowering Ω_{bond} via π^* -backdonation. This correlation is consistent with the dynamical framework and does not imply that dynamical optimization was the historical design intent. The path toward ambient nitrogen fixation at lower temperature and pressure requires a material where $\chi_{\text{TS}} \approx 1$ is achieved without those process conditions.

Recent first-principles calculations on ferroelectric $\text{Cu}_n(\text{CrSe}_2)_{n+1}$ monolayers demonstrate that reversible polarization switching dynamically decouples hydrogen adsorption and desorption, reducing the Heyrovsky barrier from 0.71 eV to 0.36 eV (Tafel: 1.29 eV to 0.72 eV) while incurring a switching cost of only 0.32 eV²⁷. Such cycling between electronically distinct adsorption- and desorption-favorable states illustrates an alternative to static binding optimization, aligning with the dynamical decoupling strategy discussed here. Such polarization switching dynamically modulates Γ_{eff} , demonstrating that χ can be tuned not only by static promoter loading but also by dynamic switching between electronically distinct surface states.

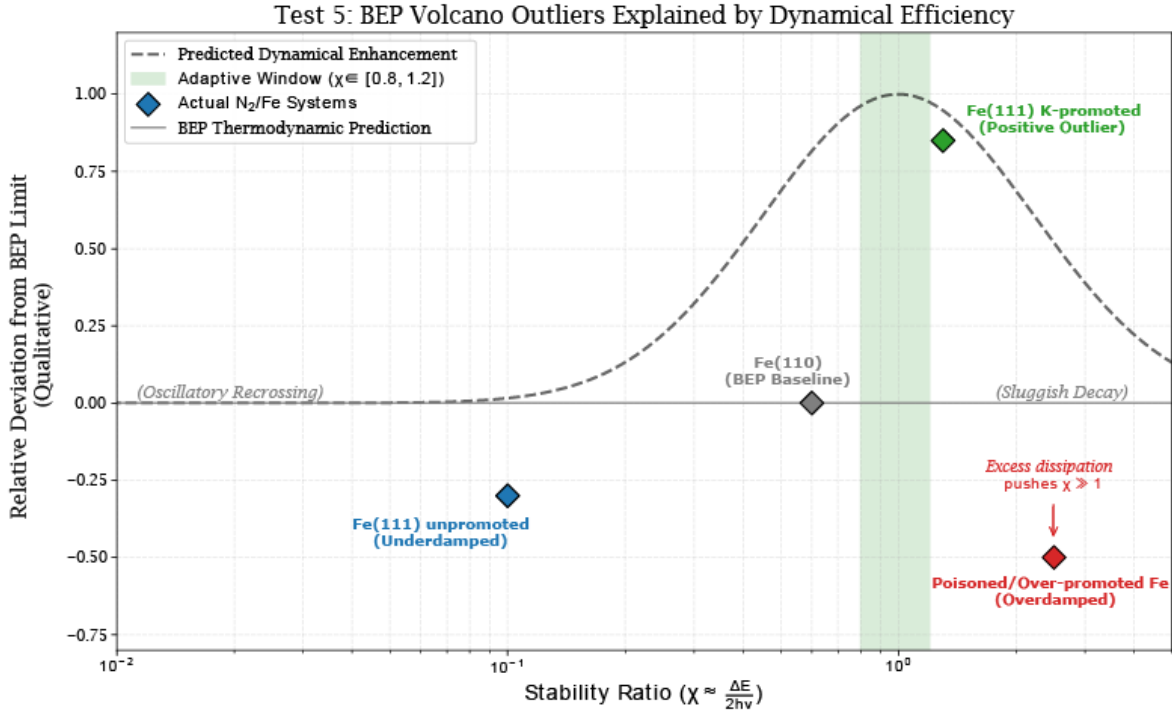


Figure 2: BEP volcano outliers explained by dynamical efficiency (Test 5). Damping ratio $\chi \approx \gamma_{\text{NN}}/(2\hbar c\tilde{\nu})$ (Route C proxy) on the horizontal axis (log scale); deviation from the BEP thermodynamic prediction on the vertical axis. Fe(110) sits at the BEP baseline ($\chi \approx 0.61$; thermodynamics adequate, dynamics partially limiting). Fe(111) unpromoted is a negative outlier ($\chi \approx 0.10$; oscillatory recrossing suppresses activity beyond the thermodynamic expectation). Fe(111) K-promoted is a positive outlier ($\chi \approx 1.28$; near-critical, minimal recrossing). Over-promoted Fe lies in the overdamped regime ($\chi \gg 1$; sluggish commitment reduces TOF below the BEP baseline). Dashed curve: predicted dynamical enhancement profile from this framework.

For catalytic systems the relevant χ governing turnover is the approach-damping ratio, defined as the value of χ in the precursor well as the coordinate approaches the barrier, not the TS value χ_{TS} at the saddle point itself. In the PGH formulation⁶, the turnover maximum is set by the approach-region friction: the ground-state DFT estimates in Table 8 describe this regime, making χ_{GS} the operationally correct quantity for screening. Although χ_{TS} is substantially larger (Supplementary §3.3 brackets $\chi_{\text{TS}} \in [2, 7]$ for K-promoted Fe(111) using the Nosir et al. potential-energy surface²⁶), it is the approach dynamics that

determine whether the coordinate commits to the barrier or recrosses.

Constrained χ -space and the origin of domain-specific optima

The standard-state periodic lattice χ map (Figure 3), where $\chi_{\text{lattice}} = \pi\lambda_{ep}T/(2\Theta_D)$ derives from the Debye-model electron-phonon coupling rate at the bath frequency $\Omega_{\text{bath}} = k_B\Theta_D/\hbar$ ^{22,23}, reveals a structural fact about which systems can access both sides of the $\chi = 1$ boundary. In electron-transfer and PCET systems, both dynamical phases are accessible by varying solvent or composition, and efficiency maxima emerge near the phase boundary for structural reasons, not coincidence. Transition-metal surfaces are categorically different: electronic friction at the Fermi level is intrinsically large, locking these systems into the overdamped sector regardless of promoter loading or temperature within the range of practical operation. There is no underdamped branch to compete with and no Kramers turnover in the kinetic sense; catalytic optimization on metals occurs entirely within the overdamped regime.

This produces a coherent classification (Table 9). The separatrix $\chi = 1$ is a generator-level boundary common to all domains considered here; the occupancy of χ -space is substrate-determined. When a system is structurally confined to one dynamical phase, optimization reorganizes internally to that phase. This is not an exception to the framework; it is the framework applied to constrained systems.

Domain	Accessible χ -space	Location of optimum
ET / PCET, solvent dynamics	Both phases	Near separatrix ($\chi \approx 0.8$ – 1.2)
Metallic surface catalysis	Overdamped only	Internal to overdamped regime; optimal relative to other metals
Wide-gap insulators, noble gases	Underdamped only	Dissipation is not the kinetic bottleneck

Table 9: Accessible χ -space and optimum location by domain.

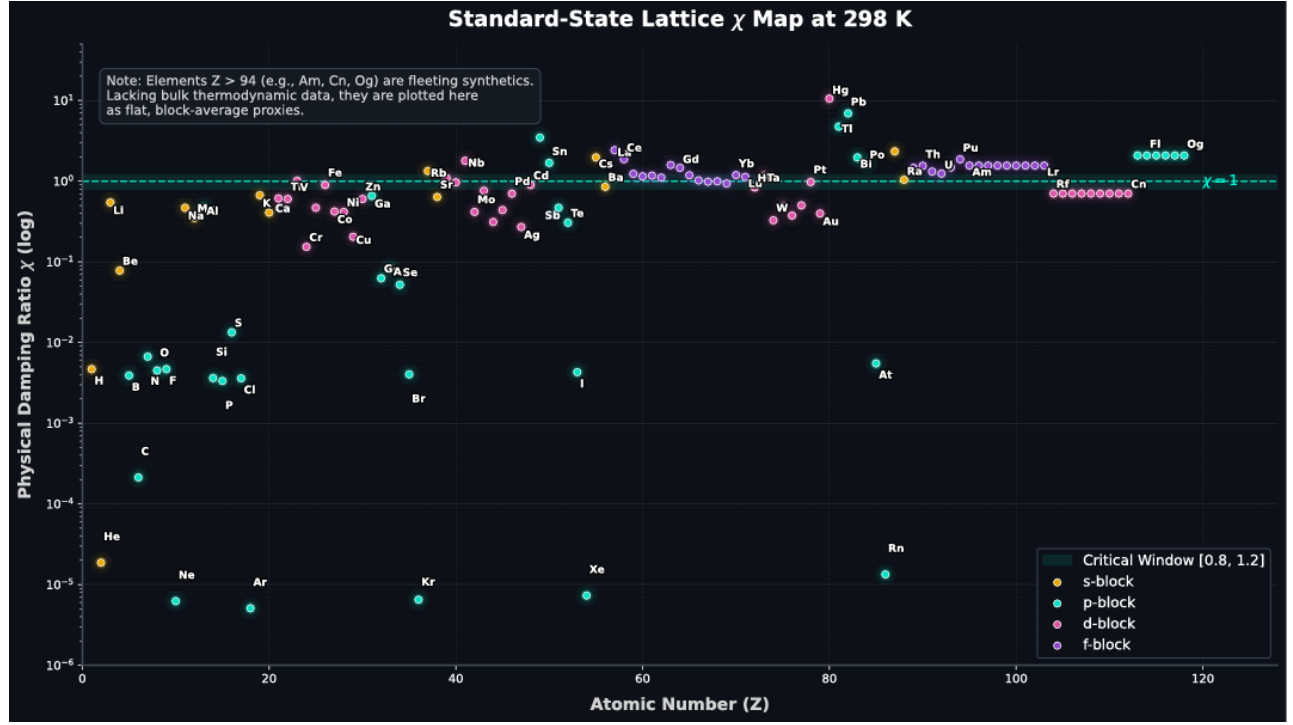


Figure 3: Standard-state lattice χ map at 298 K. $\chi_{\text{lattice}} = \pi\lambda_{ep}T/(2\Theta_D)$ is the Debye-model electron-phonon damping ratio: the damping ratio $\Gamma_{\text{eff}}/(2\Omega)$ evaluated at the bulk phonon bath frequency. χ_{lattice} is not χ_{TS} ; it is a substrate-damping proxy that describes how strongly the phonon bath couples to an electronic reactive coordinate at the Fermi level, and therefore which region of χ -space a material inherits as its operating environment. Effective catalytic transition metals cluster near $\chi \approx 1$; wide-gap insulators and noble gases occupy the underdamped floor; heavy-atom metals (Hg, Pb, Tl) are strongly overdamped. The map uses λ_{ep} (electron-phonon coupling) rather than phonon self-damping: the ratio χ measures how an electronic degree of freedom is damped by the phonon bath, not how phonons damp themselves; using phonon self-damping would place Fe at $\chi \approx 0.01\text{--}0.08$ and remove all near-critical transition metals. The clustering near $\chi \approx 1$ emerges entirely from independently tabulated λ_{ep} and Θ_D (McMillan 1968; Allen–Dynes 1975; Kittel 8th ed.) and is not imposed by parameter selection. Standard allotrope at STP; elements $Z > 94$ use block-average proxies (no bulk thermodynamic data available for synthetics).

5 Validity Boundaries: When the Framework Does Not Apply

This framework describes classical, thermally activated barrier crossing in a near-Markovian frictional environment. Table 10 gives the conditions required for valid use and those that exclude the framework.

Framework applies	Framework does not apply
Thermally activated, over-barrier reaction ($k_B T \gtrsim \hbar \Omega$)	Tunneling-dominated: $\text{KIE} > \sim 7$, $T \ll \hbar \Omega / k_B$
Near-Markovian bath ($\Omega \tau_{\text{mem}} \lesssim 1$)	Strongly non-Markovian bath ($\Omega \tau_{\text{mem}} \gg 1$)
Identified dominant reactive coordinate, harmonic near TS	Strongly anharmonic or multi-mode entangled reactive coordinate
Near-adiabatic: moderate electronic coupling H_{ab}	Strictly nonadiabatic: $H_{ab} \rightarrow 0$ (friction regime never reached)
Lorentzian fraction $> 50\%$ in IR (Route A)	Inhomogeneous-dominated spectra; raw FWHM at high temperature

Table 10: Validity conditions for the framework.

The tunneling boundary is the most commonly encountered exclusion condition. A KIE greater than approximately 7 for H/D indicates substantial hydrogen tunneling contribution; the χ framework describes the classical component only, and instanton or open-quantum-systems methods are required for the tunneling component. For systems near this boundary, the classical and tunneling contributions can be separated by their distinct temperature dependences before applying this analysis to the classical component.

When $\tau_{\text{mem}} \sim 1/\Omega_{\text{TS}}$, the Grote–Hynes ⁷ frequency-dependent friction applies and the effective damping becomes: $\Gamma_{\text{eff}}(\Omega) \approx \gamma_0(1 + (\Omega\tau_{\text{mem}})^2)^{-1/2}$. The $\chi = 1$ boundary broadens into a band, but the boundary persists. For $\Omega\tau_{\text{mem}} \lesssim 0.3$, the Markovian approximation is adequate. For $\Omega\tau_{\text{mem}} \sim 1$, the target window should be widened to $[0.5, 1.5]$ to account for memory-induced broadening.

6 Summary of Estimator Equations and Decision Criteria

Estimator routes and regime thresholds are collected in Table 11 for reference. The cross-domain clustering test (Test 8, Supplementary §4) requires independent systems: one representative optimum per independent chemical system per domain. Multiple solvents from the same donor–acceptor pair do not count as independent; multiple promoter loadings on the same surface do not count as independent. The minimum valid dataset for Test 8 is one ET system at its optimal solvent, one PCET system at its optimal pressure or composition, and one catalytic system at its optimal promoter loading. Inflating N by counting within-series replicates as independent systems invalidates the cross-domain test. The log-uniform prior assumed in the Bayesian clustering analysis (Supplementary §4, Test 8) does not affect the qualitative clustering conclusion; alternative prior choices shift the resulting p -value by $\mathcal{O}(1)$ factors only. The log-uniform prior on $[10^{-4}, 10^1]$ assumes no prior knowledge of where optima should fall; this is the most conservative choice, and any prior that does not artificially concentrate probability near $[0.8, 1.2]$ yields the same qualitative conclusion.

Estimator	Equation	Validity condition
Route A (spectroscopic)	$\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$	Lorentzian fraction > 50%; coverage-independent linewidth
Route B (solvent relaxation, relative)	$\chi_{\text{ET}}^{(A)}/\chi_{\text{ET}}^{(B)} = \tau_L^{(A)}/\tau_L^{(B)}$	Preferred; m_{eff} cancels. Absolute: compute full bracket over $m_{\text{eff}} \in [50, 500]$ amu
Route C (DFT friction)	$\chi_{\text{DFT}} = (\Gamma_{\text{int}} + \gamma_{NN})/(2hc\tilde{\nu})$	Mean-field DFT valid; treat result as lower bound on χ_{TS}
Regime thresholds: $\chi < 0.8$ underdamped; 0.8–1.2 near-critical (optimal); $\chi > 1.2$ overdamped Framework not applicable when: KIE > 7; Gaussian > Lorentzian in IR; $\Omega\tau_{\text{mem}} > 1$; $H_{ab} \rightarrow 0$		

Table 11: Summary of estimator equations, validity conditions, and regime thresholds.

Author contributions

N. Christensen: Conceptualization, Methodology, Formal Analysis, Investigation, Writing, Review & Editing, Visualization.

Conflicts of interest

There are no conflicts to declare.

Data availability

All numerical data and Python code supporting this work are deposited at Zenodo (10.5281/zenodo.17891562) and GitHub (<https://github.com/SymCUniverse/Chemistry>). The computational implementation is also provided in the Supplementary Information. References cited in the Supplementary Information are included in the reference list of this article.

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